

AN IDENTIFYING CODE OF DENSITY 59/156 IN THE 6-ROW SQUARE STRIP

ABSTRACT. Let S_k be the subgraph of the square lattice induced by $\mathbb{Z} \times \{0, \dots, k-1\}$ and let $d^*(S_k)$ be the minimum density of an identifying code of S_k for radius 1. Theorem 30 of the Lobstein–Hudry–Charon survey records $d^*(S_k)$ for $k \leq 5$ and, for $k \geq 6$, the bracket $7/20 + 1/(20k) \leq d^*(S_k) \leq 7/20 + 3/(10k)$, which at $k = 6$ is $[43/120, 2/5]$ of width $1/24$. We exhibit an explicit period-52 identifying code of S_6 of density $59/156$, printed here in full, giving

$$d^*(S_6) \leq \frac{59}{156} = 0.3782051282\dots < \frac{2}{5},$$

so the recorded interval at $k = 6$ narrows from above; nothing in the source is contradicted. Verifying the code from the definition is a finite check of 312 vertices and 1612 vertex pairs and needs no solver; the accompanying program carries it out. Only the upper bound is established here; the exact value of $d^*(S_6)$ is left open, and no claim is made that $59/156$ is close to it.

1. THE SOURCE STATEMENT

Let S_k denote the subgraph of the square lattice induced by $\mathbb{Z} \times \{0, \dots, k-1\}$, the k -row *infinite square strip*. For a vertex v write $N[v]$ for its *closed* neighbourhood in S_k . A set $C \subseteq V(S_k)$ is an *identifying code* (of radius 1) if

- (1) $N[u] \cap C \neq \emptyset$ for every u , $N[u] \cap C \neq N[v] \cap C$ for all $u \neq v$.

The set $N[u] \cap C$ is the *identifier* of u . We take the *density* of C and the invariant d^* from the source, so that every statement below is about exactly the quantity Theorem 30 brackets: [5, Section 6.4.1] defines the density of a code C in $G_\pi^{[k]}$, for every $k \geq 1$, as $\limsup_{n \rightarrow \infty} |C \cap N_n[v]|/|N_n[v]|$, where $N_n[v]$ is the ball of radius n about an arbitrary vertex v , and $d^*(S_k) = \partial_1^{Id}(G_S^{[k]})$ is the infimum of the densities of the identifying codes of S_k .

The code exhibited in Section 2 is periodic with period 52, so its density is a limit rather than merely a limsup, and it takes the same value if the balls $N_n[v]$ are replaced by rectangles $[-n, n] \times \{0, \dots, k-1\}$: at $k = 6$ the ball $N_n[v]$ and the rectangle $[-n, n] \times \{0, \dots, 5\}$ agree except on at most $6 \cdot 6 = 36$ cells near the two ends, out of the $6(2n+1)$ cells of the rectangle, so the two window ratios differ by $O(1/n)$ and have the same

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limit. Either window therefore yields the density computed in Section 2, which the program evaluates as the exact rational 118/312 over one period.

The state of knowledge is recorded in A. Lobstein, O. Hudry and I. Charon, *Locating-Domination and Identification*, Chapter 6 of *Topics in Domination in Graphs* [5], Section 6.4.1.1 “The Square Grid”, where ∂_1^{Id} is that chapter’s symbol for the minimum density we call d^* and the superscript $[k]$ denotes the strip of height k . Its Theorem 30 reads, in part (with the source’s stacked fractions written inline and parentheses added in $1/(20k)$ and $3/(10k)$):

Theorem 30 (f) ([23]) For $k \geq 6$, we have

$$7/20 + 1/(20k) \leq \partial_1^{Id}(G_S^{[k]}) \leq 7/20 + 3/(10k).$$

Parts (a)–(e) record exact values for $k = 1, \dots, 5$; for $k \geq 6$ the source records only the bracket of part (f). At $k = 6$ that bracket is

$$\left[\frac{7}{20} + \frac{1}{120}, \frac{7}{20} + \frac{3}{60} \right] = \left[\frac{43}{120}, \frac{2}{5} \right], \quad \frac{2}{5} - \frac{43}{120} = \frac{1}{24},$$

the upper endpoint being exactly $2/5$. Part (f) is a proved theorem carrying a gap at every $k \geq 6$; the source leaves the value at $k = 6$ undetermined, recording only the interval $[43/120, 2/5]$. What the present note does is narrow that recorded interval from above.

Attribution. The bracket endpoints of Theorem 30(f) are due to Bouznif, Havet and Preissmann [2]. Since the transfer-matrix and periodicity machinery of [4], [2] and [1] could in principle evaluate $k = 6$, we record what each of the works we read reaches: [3] gives $k = 1, 2$ and the first $k \geq 6$ bracket; [2] proves the two bracket endpoints and, as an exact value, only $d^*(S_3) = 7/18$; [4] supplies the periodicity theorem and the minimum-cycle-mean route and evaluates $k = 4$ and $k = 5$; [1] resolves heights 1, 2, 3, 4. None of them states a value or a code at $k = 6$, which is why [5] still records that cell as an interval in 2020. No priority claim is made beyond that: no search was carried out past the works listed here, and 59/156 or a better bound may well be recorded elsewhere. What is offered is the code of Section 2 and the verification of (1) for it.

2. THE OBJECT, AND THE UPPER BOUND

Let $L = 52$, let the 52 integers below be $c_0, \dots, c_{51}$, read left to right and top to bottom, and put

$$C = \{(x, y) \in \mathbb{Z} \times \{0, \dots, 5\} : \text{bit } y \text{ of } c_{x \bmod 52} \text{ equals } 1\}.$$

```

18 24 19 2 58 2 41 10 40 3 58 0 21 21 21 0 58 3
40 10 41 2 58 2 22 16 51 2 26 16 23 16 37 20 5 48
23 0 42 41 42 0 23 48 5 20 37 16 23 16 50 6

```

Every c_x satisfies $0 \leq c_x < 64$. Equivalently, writing 1 for a cell in C and a dot for a cell outside it, one period of C is the following 6×52 array, row y being the row y of the strip:

```

row 0  .1...1..1..111..1..1.....1...1.1.1.1..1.1.1.1...
row 1  1.1111.1.11.....11.1.1111.111.1.....1.1.1.1.....1.11
row 2  .....111.....1.....1.111.1.....1.111.1..1
row 3  .1..1.111.1.....1.111.1.....1.....111.....
row 4  111.1.....1.111.1.....1.111.1111.1.11.....11.1.1111.
row 5  ....1.1.1.1.....1.1.1.1...1.....1..1..111..1..1...1.
    
```

The row weights are 18, 26, 15, 15, 27, 17; they sum to 118 code cells out of the $6 \cdot 52 = 312$ cells of the fundamental domain.

Theorem 1. *The set C above is an identifying code of S_6 , of density $118/312 = 59/156$, so*

$$d^*(S_6) \leq \frac{59}{156} < \frac{2}{5}, \quad \frac{2}{5} - \frac{59}{156} = \frac{17}{780}.$$

Proof. C is invariant under $(x, y) \mapsto (x + 52, y)$ and meets a fundamental domain in 118 of 312 cells, so its density is $118/312 = 59/156$. It remains to verify (1), and the point is that this is a *finite* check with an explicit bound.

If $N[u] \cap N[v] \neq \emptyset$ then some vertex is at distance at most 1 from both u and v , so $d(u, v) \leq 2$. Contrapositively, if $d(u, v) \geq 3$ the two closed neighbourhoods are disjoint; if moreover both identifiers are nonempty — which is the domination clause — then they are distinct sets, since a nonempty set cannot equal a set disjoint from it. So the separation clause of (1) only has to be verified for pairs at distance at most 2, and by the 52-periodicity of C only for pairs whose left-most vertex lies in $[0, 51] \times \{0, \dots, 5\}$.

That is 312 domination tests and 1612 pair tests, each one an inspection of at most 10 cells of the printed array; all of them pass. The accompanying program carries out this verification and reports no failures. $\square$

3. SCOPE

59/156 lies strictly inside $[43/120, 2/5]$, with $2/5 - 59/156 = 17/780$ and $59/156 - 43/120 = 31/1560$: Theorem 1 narrows from above the interval the source records at $k = 6$; nothing in Theorem 30(f) is contradicted. The value of $d^*(S_6)$, and the cases $k \geq 7$, are left open here.

The tile of Section 2 was found by a randomised local search over periodic patterns of period dividing 52, and 52 was simply the period at which a valid pattern of this weight turned up; no exhaustive search was performed at that or any other period, and no certified minimum is claimed. In particular we make no claim that 59/156 is close to optimal: it is one interior point of the recorded interval $[43/120, 2/5]$, and the transfer-matrix and minimum-cycle-mean machinery of [4], [2] and [1] may well settle $k = 6$ outright. Only the inequality $d^*(S_6) \leq 59/156$ is asserted.

4. VERIFICATION

The accompanying program `verify.py` (Python 3.9+, standard library only, no external data file, exact integer and `Fraction` arithmetic) reads the 52 masks printed in Section 2, regenerates the six row strings from them, and

re-derives the row weights, the count 118 of 312 and the density $59/156$ as an exact rational, and then verifies (1): no domination failures over the 312 vertices of the fundamental domain and no separation failures among the relevant pairs. Its recorded run is `verify.output.txt`, one `PASS` line per check. The run also prints some checks corresponding to material deleted from this note (the empty columns, the minimality of the period, the bracket arithmetic, and a sweep of pairs at distance 3 to 6); those support nothing claimed here. It establishes the upper bound of Theorem 1 and nothing else.

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